FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE **JEE(Advanced)-2016**

ANSWERS, HINTS & SOLUTIONS

HALF COURCE TEST –VI
(Paper - 1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	D	A
2.	C	C	A
3.	C	A	A
4.	B	C	A
5.	A, B, C, D	A, B, D	B, C
6.	A, B, D	B, C, D	A, C
7.	A, C	B, C, D	A, B, C
8.	A, B	A, B	A, B
9.	A	A	B
10.	A	B	A
11.	B	C	C
1.	(A) → (r) (B) → (p) (C) → (s) (D) → (q)	(A) → (p) (B) → (s) (C) → (q, t) (D) → (r)	(A) → (s) (B) → (p) (C) → (r) (D) → (q)
2.	(A) → (p, q, r) (B) → (s, t) (C) → (s, t) (D) → (p, q, r)	(A) → (r); (B) → (p, q, t) (C) → (p, q, s) (D) → (p, t)	(A) → (p) (B) → (q) (C) → (r) (D) → (s)
1.	4	2	8
2.	2	2	5
3.	3	6	1
4.	1	3	8
5.	4	4	7
6.	4	9	7

Physics

PART – I

SECTION – A

1. $V_1 = 4.5$ m/s, $V_2 = 3$ m/s

2. $F + \rho g(L - x) - \rho gL = \frac{d}{dt}(\rho xu)$

3. $v_{p_x} = 5$, $v_{p_y} = 5$

   $5\sqrt{2}$ = 7.1 m/s approx.

4. Solve the problem using μ as the coefficient of kinetic friction. Linear velocity radius to zero before angular velocity becomes zero.

5. $y = A \sin(kx - \omega t + \phi)$
   at $t = 0$, $x = 0$, $y = 5$ mm
   at $t = \frac{1}{24}$, $x = 0$, $y = 0$
   also $\lambda = 2m$, $A = 10$ mm
   Solve to get answer.

6. Given PT = Constant and PV = nRT
   $\Rightarrow$ P^2V = constant
   $\Rightarrow$ $PV^{1/2} = K$.
   From first law of thermodynamics
   $\Delta Q = \Delta U + \Delta O$

   $$C\Delta T = C_V T + \left(\frac{P_f V_f - P_i V_i}{1 - \frac{1}{2}} \right)$$

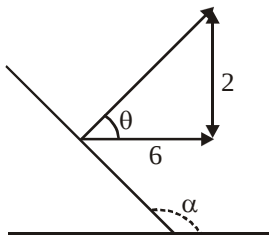
   $C = C_V + 2R$

   $$33.24 = \frac{R}{Y - 1} + 2R$$

   $\Rightarrow$ $Y = 1.5$ and $Y = 1 + \frac{2}{f}$

   $\therefore$ $f = 4$.

7. Impulse = change in momentum
   $= 2(\vec{v}_2 - \vec{v}_1)$
   $= 2(3\hat{i} + \hat{j})$
   As impulse is in the normal direction of colliding surface

   $\tan\theta = \frac{1}{3}$

   $\theta = \tan^{-1}\left(\frac{1}{3}\right)$

   $\alpha = 90° + \tan^{-1}\left(\frac{1}{3}\right)$.

8. $H = \dfrac{KAd\theta}{dr} = K(2\pi rL)\dfrac{d\theta}{dr}$

$\therefore \quad \int_{r_1}^{r_2} \dfrac{dr}{r} = \dfrac{2\pi KL}{H}\int_{\theta_1}^{\theta_2} d\theta$

$\dfrac{dQ}{dt} = H = \dfrac{2\pi KL(\theta_2 - \theta_1)}{\ln\left(\dfrac{r_2}{r_1}\right)} = 80\pi$

$\Rightarrow \quad \dfrac{dm}{dt}L = 80\pi$

$\Rightarrow \quad \dfrac{dm}{dt} = \dfrac{80\pi}{L} = \dfrac{80\pi}{80\times 4200} = \dfrac{\pi}{4200}$ kg/second.

9. Tensile stress $= \dfrac{F\cos\theta}{A/\cos\theta} = \dfrac{F\cos^2\theta}{A}$

10. Tensile stress is maximum when $\cos^2\theta$ is maximum i.e. at $\theta = 0^o$

11. Shearing stress $= \dfrac{F\sin\theta}{A/\cos\theta} = \dfrac{F\sin\theta\cos\theta}{A} = \dfrac{F\sin 2\theta}{2A}$

SECTION – C

1. $\psi = a\sin\left[\omega t - \dfrac{2\pi}{\lambda}(x\cos\alpha + y\cos\beta)\right]$

represents a wave travelling along a line in x-y plane through origin making an angle α with x-axis and β with the y-axis.

$\Delta\phi = \dfrac{2\pi}{\lambda}\left[(x_2 - x_1)\cos\alpha + (y_2 - y_1)\cos\beta\right]$

comparing with the given equation, we get

$\alpha = 30^o$, $\beta = 60^o$, $\lambda = 1$m, $\omega = 30$/s

Also, $\Delta\phi = 4\pi$.

2. $P = \sigma A e (3000^4 - 300^4) = 2$ kW.

3. $v = \dfrac{50}{100} V_e = \dfrac{1}{2}\sqrt{\dfrac{2GM}{R}}$

Applying energy conservation

$\Rightarrow \quad -\dfrac{GMm}{R} + \dfrac{1}{2}mv^2 = -\dfrac{GMm}{(R+h)}$

$v^2 = \dfrac{2GM}{R} - \dfrac{2GM}{R+h}$

$\Rightarrow \quad \dfrac{1}{4}.\dfrac{2GM}{R} = 2GM\left(\dfrac{1}{R} - \dfrac{1}{R+h}\right)$

$\Rightarrow \quad \dfrac{1}{4R} = \dfrac{h}{R(R+h)}$

$\Rightarrow$ R + h = 4h

$\Rightarrow$ h = R/3

4 $N = mg$

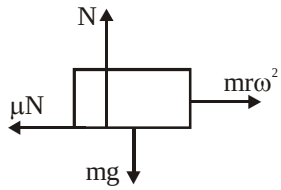
$\mu N = m r \omega^2$

$\mu mg = m.\ 2 \sin\theta\ \omega^2$

$$\omega = \sqrt{\frac{\mu g}{2\sin\theta}}$$

$$\omega = \sqrt{\frac{0.1\times 10}{2\times \sin 30}}$$

$\omega = 1$ rad / s.



5. $x = A\cos\omega t$

$$v = \frac{dx}{dt} = -A\omega\sin\omega t$$

$$a = \frac{d^2x}{dt^2} = -A\omega^2\cos\omega t$$

$$a_{max} = A\omega^2$$

$$\tau_{max} = I\,\alpha_{max} = I\frac{a_{max}}{R} = \frac{1}{2}MR^2\frac{A\omega^2}{R} = \frac{1}{2}MR\,A\omega^2$$

$\tau_{max} = 4$ N.m

6. $A_1v_1 = A_2v_2$

$v_2 = 16$ m/s

Applying Bernoulli's equation

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2 \quad (\rho = 0.9 \times 10^3 \text{ kg/m}^3)$$

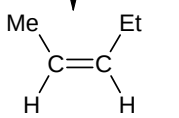
Chemistry

PART – II

SECTION – A

1. $Mg_2C_3 \xrightarrow[(X)]{H_2O} CH_3-C\equiv C-H \xrightarrow[(ii)CH_3-CH_2-Cl]{(i)\ Na} H_3C-C\equiv C-CH_2-CH_3$ (Y)

P_2 - catalyst

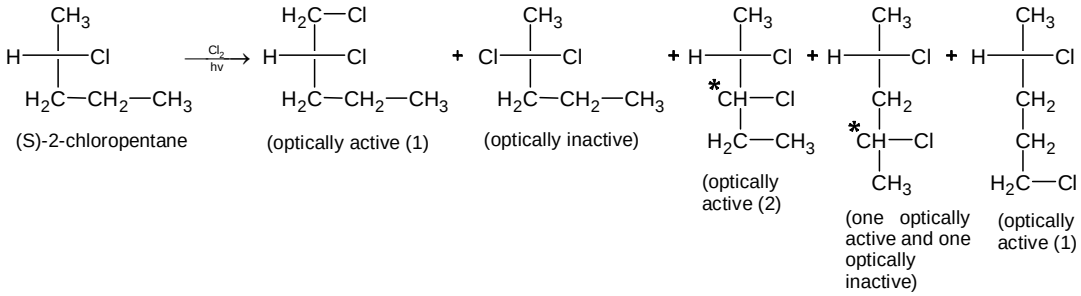


(2Z) - pent - 2 - ene

Hence (D) is correct.

2. $BCl_3 + 3NH_4Cl \xrightarrow{140^0C} \underset{\text{(A) (trichloroborazine)}}{B_3N_3H_3Cl_3} \xrightarrow[\text{(reduction)}]{NaBH_4} \underset{\text{(Borazine or Inorganic benzene)}}{B_3N_3H_6} \xrightarrow{H_2O} NH_3 + \underset{\text{(B)}}{H_3BO_3} + H_2$

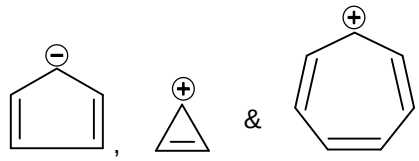
Hence (C) is correct.

3. 

Hence (A) is correct.

4. The limiting reagent here is HCl. So calculating with the limiting reagent, the total number of products is 7.5.

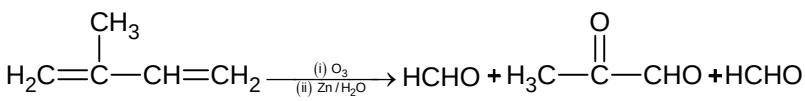
5. Symmetrical alkenes give same products.

6. 

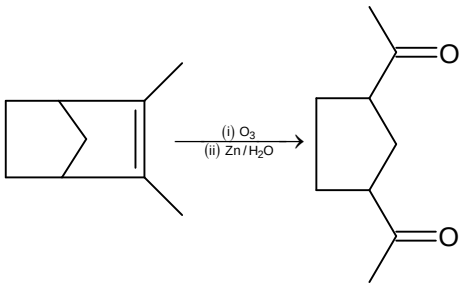
have $(4n + 2)\pi$ electrons.

7. CCl_4 does not undergo hydrolysis.

8. KE_{max} of photoelectrons ejected depends only on frequency and not on its intensity.
Also, photoelectric effect is a direct evidence for the particle nature of light not of dual nature.
Hence (A), (B) is correct.

9. 

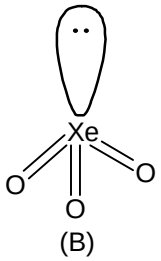
10.



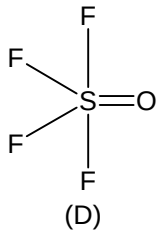
11. Oxidative ozonolysis

SECTION – B

1.



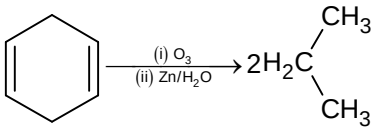
(B)



(D)

Trigonal bipyramidal

2. (A)



(B) $H_2C{=}CH{-}CH{=}CH{-}CH_3 \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} 2HCHO + \begin{matrix}CHO\\|\\CHO\end{matrix} + H_3C{-}CHO$

(C) $(H_3C)_2C{=}CH{-}CH{=}CH_2 \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} \begin{matrix}H_3C\\\diagdown\\C{=}O\\\diagup\\H_3C\end{matrix} + \begin{matrix}CHO\\|\\CHO\end{matrix} + HCHO$

(D) $H_3C{-}CH{=}CH_2 \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} H_3C{-}CHO + HCHO$

SECTION – C

1. $AgSCN \rightleftharpoons Ag^+ + SCN^-$

a a

$AgBr \rightleftharpoons Ag^+ + Br^-$

b b

$$\frac{(K_{sp})_{AgSCN}}{(K_{sp})_{AgBr}} = \frac{(a+b)a}{(a+b)b} = \frac{1.0\times10^{-12}}{5.0\times10^{-13}} = 2$$

2.

	N_2	+ $3H_2$	$\longrightarrow 2NH_3$
Initial	2	5	0
(after N_2 reacted 50%)	1	2	2

Total moles of N_2, H_2 and NH_3 = 5 mol (after reaction)
Initial mol = 7

Pressure at start = 7 atm.
Hence pressure at equilibrium = 5 atm.
Partial pressure of $NH_3 = \frac{2}{5} \times 5 = 2$ atm.

4. $\mu = 4.9$ B.M, $4.9 = \sqrt{n(n+2)}$
Number of unpaired electorn in $Mn^{x+} = 4$, x = 3

5. Volume withdrawn = $(20 \times 2) - 20 = 20$ L
Numer of balloons = $\frac{20}{5} = 4$

6. Maximum number of electron in 3rd orbit = 18
$\therefore$ electrons with spin quantum number $m_s = -1/2 = 9$

Mathematics

PART – III

SECTION – A

1. Put x = 1 and x = – 1 and adding

$$a_0 + a_2 + \ldots + a_{50} = \frac{3^{25}+1}{2} = \frac{(1+2)^{25}+1}{2}$$

$$= \frac{{}^{25}C_0 + {}^{25}C_1 \cdot 2 + {}^{25}C_2 \cdot 2^2 + \ldots + {}^{25}C_{25} \cdot 2^{25} + 1}{2}$$

$$= \frac{2\left[1 + {}^{25}C_1 + {}^{25}C_2\, 2 + \ldots + {}^{25}C_{25} 2^{24}\right]}{2}$$

$$= 2\left[13 + {}^{25}C_2 + \ldots + {}^{25}C_{25}\, 2^{25}\right] \Rightarrow \text{even.}$$

2. Circle touching y = x at (0, 0) can be taken as $x^2 + y^2 + \lambda(x - y) = 0$ … (i)
This bisects the circumference of the circle $x^2 + y^2 + 2y - 3 = 0$ … (ii)
Hence radical axis between (i) and (ii) must pass through the centre of (ii) which is (0, – 1)
Radical axis is $\lambda(x - y) - 2y + 3 = 0$, hence x = – 5 put x = – 5 in (i) to get the result.

3. Number of positive integral solution of x + y + z + w = 50 is ${}^{50}C_3$
Number of positive integral solution of x + y + z + w = 19 is ${}^{19}C_3$

4. For equilateral triangle

$$|z_2 - z_1| = |z_3 - z_2| = |z_1 - z_3| = a$$

Also $\left|\frac{z_2 + z_3}{2} - z_1\right|$ will be the length of median as well as altitude i.e. $\left|\frac{z_2 + z_3}{2} - z_1\right| = \frac{a\sqrt{3}}{2}$

Hence, $\sin\theta = \sqrt{\frac{|z_2 - z_1|^2 + |z_3 - z_1|^2}{|z_2 + z_3 - 2z_1|^2 + |z_3 - z_2|^2}} = \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4}$.

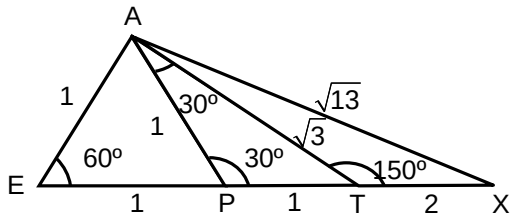
5. Any conic having its foci same as the given ellipse will have equation $\frac{x^2}{9+\lambda} + \frac{y^2}{4+\lambda} = 1$, x is variable now if the curve passes through (3, 2), we get

$\lambda = \pm 6$

Putting value of λ, we get the required answer.

6. For real roots $4a^2 - 4b(a - 1) \geq 0$
$\Rightarrow \quad a^2 - ba + b \geq 0$
Now for above inequality to hold for all values of a,
$b^2 - 4b \leq 0$ or $0 \leq b \leq 4$.

7. We have AE = EP = AP = 1 $\Rightarrow$ AP = PT = 1
$\Rightarrow \quad \Delta APT$ is isosceles
This makes ΔAET a 30º – 60º – 90º triangle,
So AT = $\sqrt{3}$ and $\angle ATX = 150^\circ$
since TX = 2
$\therefore$ By applying cosine rule in ΔATX, we get
$(AX)^2 = 3 + 4 - 4\sqrt{3}\cos 150^\circ$
$= 7 + 6 = 13$
$\Rightarrow \quad AX = \sqrt{13}$
$\therefore \quad \cos\angle XAE = \frac{13 + 1 - 16}{(2)\left(\sqrt{13}\right)(1)} = \frac{-2}{2\sqrt{13}} = \frac{-1}{\sqrt{13}}$.



8. Let A be (h, k), then altitude = k
Also AB : AC = 2 : 1, hence $AB^2 = 4AC^2$
Now, $AB^2 = h^2 + k^2$ and $AC^2 = (6 - h)^2 + k^2$

$\Rightarrow \quad 4AC^2 - \left(\frac{AC^2 + 12}{4}\right)^2 > 10$

$\Rightarrow \quad 20 - \sqrt{96} < AC < 20 + \sqrt{96}$

9-11. Slope of $AB = \frac{5}{7}$, hence slope of $PQ = \frac{-7}{5}$

Let equation of PQ be 7x + 5y = C, then coordinates of P & Q will be $\left(\frac{C}{7}, 0\right) \& \left(0, \frac{C}{5}\right)$. Now equation of AQ and BP will be $\frac{x}{7} + \frac{5y}{c} = 0$ & $\frac{7x}{C} - \frac{y}{5} = 1$

Eliminating C from these two equations gives the locus of R as the circle $x^2 + y^2 - 7x + 5y = 0$.

Now, T will be the point on the diameter through origin of this circle at a distance $\sqrt{74}$ i.e. distance of centre from origin + radius. Hence coordinate of T is (7, –5). Equation of the requires parabola $(y + 5)^2 = 2(x - 7)$. The line y = - 3 meets the parabola at (9, - 3). As the line is parallel to the axis so any ray incident along it will pass through focus $\left(\frac{15}{2}, -5\right)$ of the parabola after getting reflected. Hence the reflected ray will be travelling along the line joining (9, –3) & $\left(\frac{15}{2}, -5\right)$.

Hence equation of reflected ray will be 4x – 3y = 45.

SECTION – B

1. (A) Required numbers of ways = $^2C_1 \times {}^2C_1 \times 7! \times 2!$
(B) Required numbers of ways = 8!
(C) Required numbers of ways = $6! \times 2! \times {}^2C_1 \times {}^2C_1$
(D) Required numbers of ways = $9! \times {}^8C_2 \times {}^6C_2 \times {}^4C_2 \times {}^2C_2$

2. (A) 2 sinx, sin 2x & 2cosx are in A.P., hence sin 2x = sin x + cos x
Now, let sin x + cos x = t, then sin 2x = $t^2 - 1$, then the above equation gives $t^2 - t - 1 = 0$ or
$t = \sin x + \cos x = \frac{1 - \sqrt{5}}{2}$

(B) Let C be the sides angles, then angles will be
$\frac{\pi}{3} - \frac{c}{3}, \frac{\pi}{3}, \frac{\pi}{3} + \frac{c}{3}$

$\Rightarrow \frac{\pi}{3} + \frac{c}{3} = c \Rightarrow c = \frac{\pi}{2}$, hence angles are $\frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}$.

Now $\frac{a}{\sin\frac{\pi}{6}} = \frac{b}{\sin\frac{\pi}{2}} = \frac{c}{\sin\frac{\pi}{2}} \Rightarrow \frac{a}{1} = \frac{b}{\sqrt{3}} = \frac{c}{2}$

Hence angles are $\frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}$.

Now $\frac{a}{\sin\frac{\pi}{6}} = \frac{b}{\sin\frac{\pi}{3}} = \frac{c}{\sin\frac{\pi}{2}}$

$\Rightarrow \frac{a}{1} = \frac{b}{\sqrt{3}} = \frac{c}{2}$.

(C) Let AB = a, BC = $\sqrt{3}$ a, CA = 3a & AD = $3\sqrt{3}$ a

Now cos A + cos C = 0 $\Rightarrow \frac{a^2 + 27a^2 - BD^2}{3\sqrt{3}a^2} + \frac{3a^2 + 9a^2 - BD^2}{3\sqrt{3}a^2} = 0$

$BD^2 = 20a^2 \Rightarrow \frac{BD}{AB} = \sqrt{20} = 2\sqrt{5}$

(D) $a + c = 2b$ & $b^2 = \frac{2a^2c^2}{a^2 + c^2} \Rightarrow (a + c)^2 = \frac{8a^2c^2}{a^2 + c^2}$

$\Rightarrow (a + c)^4 - 2ac(a + c)^2 - 8a^2c^2 = 0$

$\frac{(a+c)^2}{ac} = -2, 4$

But 4 gives a = c, hence taking – 2

$a^2 + 4ac + c^2 = 0 \Rightarrow \frac{a}{c} = \sqrt{3} - 2$.

SECTION – C

1. $\frac{R}{r} = \frac{3}{1.5} = 2 \Rightarrow \Delta ABC$ must be equilateral

so $a = b = c = 2R \sin\frac{\pi}{3} = R\sqrt{3}$

Now $a\cot^2 A + b^2 \cot^3 B + c^3 \cot^4 C = R\sqrt{3}\left(\frac{1}{\sqrt{3}}\right)^2 + \left(R\sqrt{3}\right)^2\left(\frac{1}{\sqrt{3}}\right)^3 + \left(R\sqrt{3}\right)^3\left(\frac{1}{\sqrt{3}}\right)^4$

$= \frac{R}{\sqrt{3}} + \frac{R^2}{\sqrt{3}} + \frac{R^3}{\sqrt{3}} = \frac{3 + 3^2 + 3^3}{\sqrt{3}} = \frac{39}{\sqrt{3}} = 13\sqrt{3}$

$m = 13, n = 3 \Rightarrow \frac{m+n}{2} = \frac{16}{2} = 8$.

2. $\frac{2^{10}\sin\left(\frac{5\pi}{2^{11}}\right)\cos\left(\frac{5\pi}{2^{11}}\right)\cos\left(\frac{5\pi}{2^{10}}\right)\cos\left(\frac{5\pi}{2^9}\right)...\cos\left(\frac{5\pi}{2^3}\right)\cos\left(\frac{5\pi}{2^2}\right)}{2^{10}}$

$= \frac{\sin\frac{5\pi}{2}}{2^{10}} = \frac{1}{1024} = 4^{-5}$

3. Let P be (h, k), then chord of contact will be $\frac{hx}{2} - \frac{ky}{3} = 1$

Homogenising equation of hyperbola by equation of chord of contact gives

$\frac{x^2}{2} - \frac{y^2}{3} = \left(\frac{hx}{2} - \frac{ky}{3} - 1\right)^2$. If these are perpendicular lines, then $\frac{1}{2} - \frac{h^2}{4} - \frac{1}{3} - \frac{k^2}{a} = 0$

hence required locus is $\frac{x^2}{4} + \frac{y^2}{9} = \frac{1}{6}$

4. For equilateral triangle $r = \frac{R}{2}$, hence R = 8

Now in radius r of ABC is the circum radius for second triangle. Hence $x_1 = \frac{r}{2} = \frac{R}{4}$

Similarly $x_2 = \frac{x_1}{2} = \frac{R}{8}$, $x_3 = \frac{R}{16}$ …

$x_1 + x_2 + x_3 + \ldots \infty = \frac{R}{4} + \frac{R}{8} + \frac{R}{16} + \ldots\infty$

Hence sum of all these radii is $\frac{\frac{R}{4}}{1-\frac{R}{2}} + \frac{R}{2} = R = 8$.

5. R.M.S. $\geq$ A.M. $\Rightarrow \sqrt{\frac{|z-3|^2 + |z|^2 + |z+3|^2}{3}} \geq \frac{|z-3| + |z| + |z+3|}{3}$

Also $|z-3|^2 + |z|^2 + |z+3|^2 = 2(|z|^2 + 9)$, hence $|z| \geq \sqrt{10} \Rightarrow a = 3$

where as $|z-3| + |z| + |z+3| \geq |z-3+2+2+3| \Rightarrow |z| \leq 4$

$\Rightarrow b = 4$.

$a + b = 7$.

6. Let us form that equation having roots $y = g(x_i)$

i.e. $y = x^2 - 2 \Rightarrow x = \sqrt{y+2}$

$$\left(\sqrt{y+2}\right)^2 + \left(\sqrt{y+2}\right)^2 + 1 = 0$$

$\Rightarrow$ $y^5 + 20y^4 + 40y^3 + 79y^2 + 74y + 23 = 0$

$\therefore$ $g(x_1)\, g(x_2) \ldots g(x_5)$ = product of roots

$= -23$

$x_1x_2x_3x_4x_5 = -1 \Rightarrow g(x_1x_2x_3x_4x_5) = -1$

$\Rightarrow$ $g(x_1)g(x_2) \ldots g(x_5) - 30g(x_1x_2 \ldots x_5) = 7$.